THE CURIOSITY CUP 2026
A Global SAS® Student Competition

The False Growth Detector: Revealing Hidden Instability in High-Growth Regions

The Outliers

ABSTRACT

The labor market is changing faster than the indicators used to measure it. Automation and AI are reshaping occupations, and regions thriving on headline job numbers may be masking fragility. San Jose, LA, Austin, Phoenix, and Miami topped national employment rankings, yet separation rates ran so high that much of the hiring replaced workers already leaving. Payroll counts registered vitality; skill structures told a different story. This study uses the pandemic as a natural experiment: can workforce data predict, before a shock hits, which regions are structurally fragile? It develops a "False Growth Detector" that reads regional skill composition, or "Skill DNA," to separate durable expansion from growth fueled by churn. Skills are the lens: what a job demands determines whether workers stay.

A gradient boosting model trained on 2019 ONET skill and work-context vectors with BLS employment and separation data builds city-level resilience profiles and tests whether they predict post-shock 2023 outcomes. Results show skill composition outperforms employment growth as a churn predictor: 114 (61%) of 187 high-growth metros are classified as False Growth, where strong hiring conceals elevated instability. Yet skill composition remains largely absent from standard labor indicators. These findings suggest it belongs among the primary metrics for assessing regional resilience.

INTRODUCTION

Las Vegas and San Jose, at opposite ends of the wage spectrum, expose the same problem. After the pandemic, Las Vegas posted over 6% employment growth while hospitality turnover approached 85% and separations outpaced new hires by thousands per month. Nevada's chief economist called it not recovery but "restructuring." San Jose tells a parallel story: tech employment sat 37,000 jobs above pre-pandemic levels, yet engineers rotated between employers so fast that gross hiring amounted to expensive backfilling, not expansion. In both cities, the employment numbers showed a thriving economy. The underlying workforce was falling apart.

You treat employment growth as proof of economic strength, but hiring volume does not equal structural resilience. In some regions, new jobs primarily replace workers who leave high-churn occupations. Growth on paper, stagnation in practice. We call this "False Growth," as distinct from "True Growth," where expansion relies on durable skill structures and lower separation risk. The difference lies in what each occupation demands from its workers. Occupations built on autonomy, complex problem solving, and analytical reasoning hold onto their workers. Those with limited decision-making and high time pressure do not. These structural demands, what we call a region's "Skill DNA," determine workforce stability in ways that employment counts cannot. With automation and AI accelerating occupational change, reading skill composition is no longer optional. This study uses Skill DNA as a leading indicator to flag instability before it surfaces in conventional employment data.

DATA

Building that indicator requires three distinct federal datasets, each answering a different question about regional workforces. The first is ONET Skills and Work Context (2019 and 2023). These databases capture what workers actually do: occupation-level skill vectors measuring cognitive, technical, and environmental attributes like critical thinking, complex problem solving, decision-making autonomy, and time pressure. ONET scores each attribute on Importance and Level scales that measure how central the skill is to the occupation and how much the job demands. This is the raw material of each occupation's Skill DNA. The second is Bureau of Labor Statistics Occupational Employment and Wage Statistics, or OEWS (2019 and 2023). This dataset captures where those workers are: metropolitan employment counts by occupation, which you use to weight the ONET skill vectors into region-level workforce profiles. The third is BLS Employment Projections. This dataset captures what happens next: the total occupational separations rate (churn rate), the share of workers who leave an occupation. This is the target variable, the instability that Skill DNA is designed to predict. Together, the three connect what jobs demand from workers, where those jobs concentrate, and how durable those workforces prove to be.

PROBLEM

The cost of ignoring skill composition is concrete. A city that hires 10,000 workers to replace 9,500 who left looks identical, on the headline, to one that created 10,000 new positions. That is the measurement gap: aggregate employment growth captures volume but says nothing about durability, retention, or whether underlying skill demands can sustain stability. Without forward-looking indicators, policymakers and firms misread fragile regions as growth successes. Workforce development funds flow to places that appear healthy, and families relocate to cities whose economic foundations are weaker than the numbers suggest. This study addresses that gap with a False Growth framework that uses regional skill composition to estimate churn risk and separate stable, skill-supported expansion from growth sustained by replacement hiring.

DATA CLEANING/VALIDATION

To build this framework, you conduct all preprocessing and modeling in **SAS® Viya® for Learners,** with Python used only for initial file ingestion and column-name standardization across the three source datasets. Because the three datasets use different occupation codes, you standardize them with Standard Occupational Classification (SOC) codes as the common key. You remove invalid, missing, and non-detailed occupation records and convert employment counts to numeric format. You filter ONET files to Importance and Level scale types and align them to SOC codes in the OEWS data. You extract separation rates from BLS Employment Projections Table 1.10 and validate full occupational coverage.

The three datasets merge through employment weighting. You weight ONET skill and work-context scores by OEWS employment counts and average them into each metro area's Skill DNA profile. BLS separation rates, weighted the same way, form the target variable: regional churn rate. Neither exists in the source data; you construct both. The final dataset contains 396 metro areas and 92 variables: 90 skill and work-context predictors (2019), one region identifier, and one churn rate target (2023). The four-year gap is by design: can pre-pandemic skill structures anticipate post-pandemic instability? Churn rates range from 9.64% to 12.99% (mean = 11.28%, SD = 0.45%), with no extreme outliers.

ANALYSIS

You run all models in **SAS® Viya® for Learners** using **PROC REG, PROC GRADBOOST**, and **PROC TTEST**. The first step is a linear regression baseline with Employment Growth (%) as the sole predictor of 2023 churn. You compute employment growth as the

percentage change in total employment between 2019 and 2023 for each MSA, ensuring comparability across regions of different sizes.

Model	Method	Prediction Error	Variance Explained
Linear Baseline – Employment growth only	PROC REG	MSE = 0.189	R^2 = 0.052
Champion – Skill DNA Gradient Boosting	PROC GRADBOOST	ASE = 0.046	R^2 = 0.768

Table 1. Model Comparison: Linear Baseline vs. Gradient Boosting champion model

The baseline yields R^2 = 0.052 and MSE = 0.1895: employment growth explains only 5.2% of churn variation. Growth alone clearly cannot capture structural labor-market dynamics, which motivates a more flexible model. Figures 5 and 6 in the appendix show the distribution across all 396 MSAs, with growth right-skewed and concentrated between 0% and 15%.

You then train a Gradient Boosting model on 2019 ONET skill and work-context features to predict 2023 churn, partitioning the 396 MSAs 70/30 into training and validation within PROC GRADBOOST. The model converges at 60 trees with a validation ASE of 0.046 and R^2 of 0.768: skill composition explains 76.8% of churn variation where employment growth explains just 5.2%. Training and validation error remain stable, and the model generalizes well with limited overfitting.

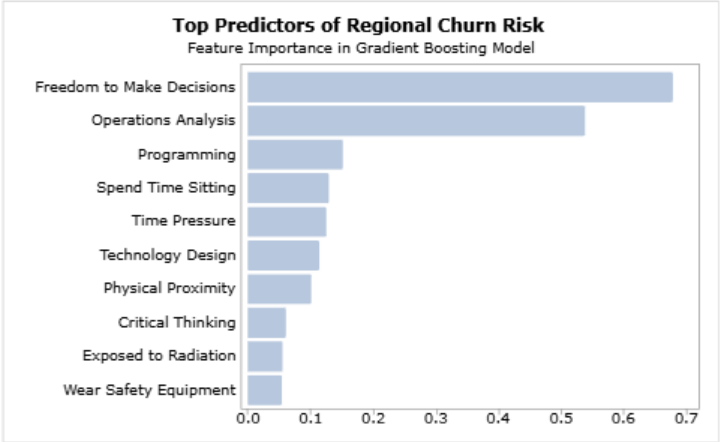


Figure 1. Top Predictors of Regional Churn Risk

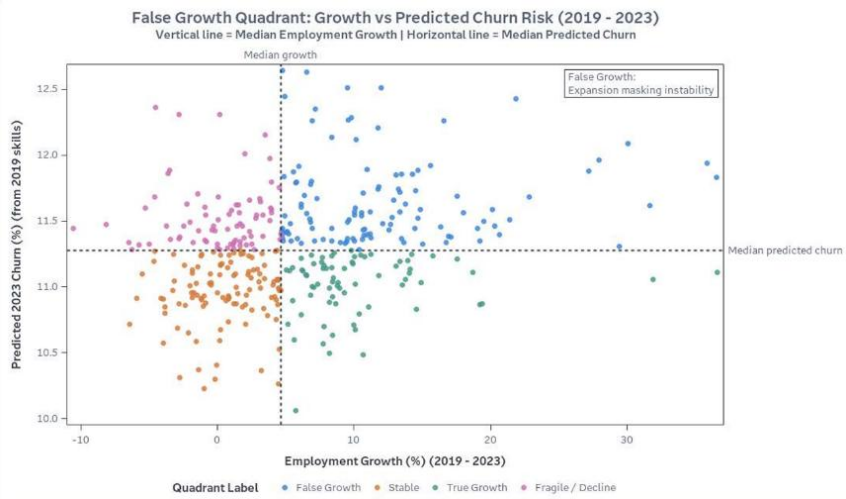
Feature importance analysis points to autonomy and cognitive complexity as the primary drivers of churn risk. Freedom to Make Decisions is the strongest predictor (importance = 0.6791), followed by Operations Analysis, Time Pressure, and Complex Problem Solving. Regions concentrated in occupations with limited autonomy and lower analytical demands face higher churn risk, even when employment is growing. Skill DNA reads the difference before employment data can.

STRUCTURAL CLASSIFICATION

The model predicts churn, but prediction alone does not prove False Growth exists in the data. You construct the False Growth quadrant by pairing predicted churn with Employment Growth (%) for all 396 MSAs. Both axes use data-driven median splits. This avoids arbitrary thresholds and produces balanced, symmetric quadrants. High churn risk is predicted churn above the median (11.26%); high employment growth is growth above the median across regions. You test quartile-based cutoffs (top and bottom 25%) as a robustness check and find consistent directional results. The final framework yields four structural categories as described in the table below.

	High Predicted Churn	Low Predicted Churn
High Employment Growth	**False Growth** Region: 114 Churn rate: 11.66%	**True Growth** Regions: 73 Churn rate: 10.99%
Low Employment Growth	**Fragile/Decline** Regions: 84 Churn rate: 11.56%	**Stable** Regions: 125 Churn rate: 10.94%

Table 2. False Growth Quadrant Definitions – Four Structural categories Based on Median Splits



You plot all 396 regions on two axes: Employment Growth (%) on the X-axis and Predicted 2023 Churn (from 2019 skills) on the Y-axis. The vertical dashed line marks median employment growth; the horizontal marks median predicted churn. The upper-right quadrant is densely populated. False Growth is not a rare exception but a common outcome among high-growth regions.

Figure 2. False Growth Quadrant: Growth vs Predicted Churn Risk

EMPIRICAL VALIDATION

The classification reveals the predicted structure. Now you test whether the original 2019 and 2023 data confirm it. You validate all four structural categories, with False Growth as the focus because it tests the core claim: growth can mask fragility. First, you compare realized 2023 churn across quadrants. False Growth regions record the highest average churn (11.66%), followed by Fragile/Decline. True Growth (10.99%) and Stable regions show materially lower churn. A PROC TTEST confirms that False Growth regions experience significantly higher realized churn than True Growth regions ($t = 13.35$, $p < 0.0001$). The fragility predicted from 2019 skill data showed up in 2023 outcomes. Second, you test the employment illusion directly. False Growth and True Growth regions grow at nearly identical rates (12.08% vs. 10.33%), yet they diverge sharply in churn. Growth alone cannot distinguish resilience from instability. Third, you assess concentration. Among all 187 high-growth MSAs, 114 (60.96%) fall into the False Growth category. In most expanding regions, growth coincides with elevated churn risk. Supporting statistics appear in Appendix Tables 3, 4, 5 and 6, with distributional diagnostics in Appendix Figures 3 and 4. All four quadrants behave differently in realized outcomes, and False Growth is a measurable structural condition, not a visual artifact.

RESULTS

The validation confirms that skill composition provides a forward-looking signal of instability that employment growth cannot capture. Skill DNA explains 76.8% of churn variation ($R^2 = 0.768$) versus 5.2% from employment growth, with 76% less prediction error (ASE = 0.046 vs. MSE = 0.1895). Of 187 high-growth MSAs, 114 (60.96%) are False Growth regions: they expand at rates comparable to True Growth regions but experience significantly higher

realized churn in 2023. These results move Skill DNA from hypothesis to operational measurement: a framework that predicts instability before it appears in employment data and separates structurally supported growth from expansion driven by replacement hiring.

SUGGESTIONS FOR FUTURE STUDIES

The framework points toward direct policy applications. Workforce agencies could integrate predicted churn into WIOA funding decisions to target reskilling investments toward False Growth regions before instability surfaces in headline employment data. That would shift funding from job-count incentives to resilience-based intervention.

On the methodological side, adding wage levels, job quality, productivity, and industry composition would sharpen the distinction between durable expansion and churn-driven growth. Extending the analysis beyond 2019–2023 would test whether False Growth represents persistent instability or temporary disruption.

Three limitations are worth noting. The median-split classification depends on thresholds, and continuous risk scoring could improve it. The analysis covers MSAs only and excludes rural labor markets. And although the churn difference is statistically significant ($p < 0.0001$), policy use should consider complementary indicators like wages and industry mix.

CONCLUSION

Regional skill composition predicts labor-market instability more effectively than employment growth alone. Among 187 high-growth regions, 114 (60.96%) are False Growth. They experience significantly higher realized churn despite similar expansion rates. Employment growth reflects scale. Skill structure reflects resilience. When the two diverge, as they do in the majority of high-growth American metro areas, the headline numbers conceal more than they reveal. As automation and AI continue reshaping occupational demands, that divergence will widen, and the cost of ignoring skill composition will grow with it.

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APPENDIX

The MEANS Procedure

Analysis Variable : Actual_Churn_2023 target_churn_2023				
Growth_Type	N Obs	Mean	Std Dev	N
False Growth	114	11.6605390	0.3537160	114
True Growth	73	10.9913971	0.3015527	73

Table 3. Mean Actual 2023 Churn by Structural Growth Category

The MEANS Procedure

Analysis Variable : Employment_Growth_Pct				
Growth_Type	N Obs	Mean	Std Dev	N
False Growth	114	12.0757456	6.7010576	114
True Growth	73	10.3337838	5.3272534	73

Table 4. Mean Employment Growth (%) by Structural Growth Category

The TTEST Procedure

Variable: Actual_Churn_2023 (target_churn_2023)

Growth_Type	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
False Growth		114	11.6605	0.3537	0.0331	10.9287	12.9913
True Growth		73	10.9914	0.3016	0.0353	9.6477	11.6626
Diff (1-2)	Pooled		0.6691	0.3344	0.0501		
Diff (1-2)	Satterthwaite		0.6691		0.0484		

Growth_Type	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
False Growth		11.6605	11.5949	11.7262	0.3537	0.3130	0.4067
True Growth		10.9914	10.9210	11.0618	0.3016	0.2593	0.3603
Diff (1-2)	Pooled	0.6691	0.5703	0.7680	0.3344	0.3035	0.3723
Diff (1-2)	Satterthwaite	0.6691	0.5736	0.7647			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	185	13.35	<.0001
Satterthwaite	Unequal	170.45	13.82	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	113	72	1.38	0.1453

Table 5. Two-Sample T-Test Results for Actual 2023 Churn: False Growth vs. True Growth

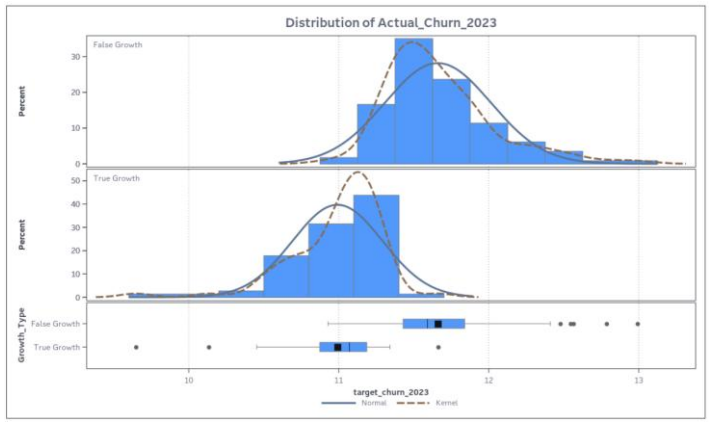

Figure 3. Distribution of Actual 2023 Churn by Growth Category

The TTEST Procedure

Variable: Employment_Growth_Pct

Growth_Type	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
False Growth		114	12.0757	6.7011	0.6276	4.6772	36.5656
True Growth		73	10.3338	5.3273	0.6235	5.0564	36.6260
Diff (1-2)	Pooled		1.7420	6.2027	0.9298		
Diff (1-2)	Satterthwaite		1.7420		0.8847		

Growth_Type	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
False Growth		12.0757	10.8323	13.3192	6.7011	5.9297	7.7049
True Growth		10.3338	9.0908	11.5767	5.3273	4.5814	6.3655
Diff (1-2)	Pooled	1.7420	-0.0924	3.5763	6.2027	5.6299	6.9062
Diff (1-2)	Satterthwaite	1.7420	-0.00396	3.4879			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	185	1.87	0.0626
Satterthwaite	Unequal	176.42	1.97	0.0505

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	113	72	1.58	0.0370

Table 6. Two-Sample T-Test Results for Employment Growth (%): False Growth vs. True Growth

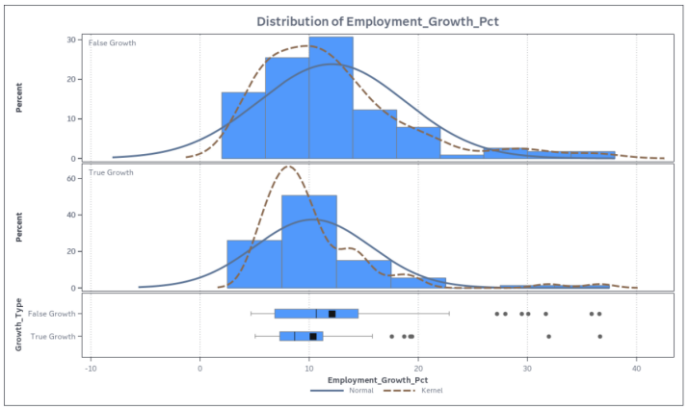

Figure 4. Distribution of Employment Growth (%) by Structural Growth Category

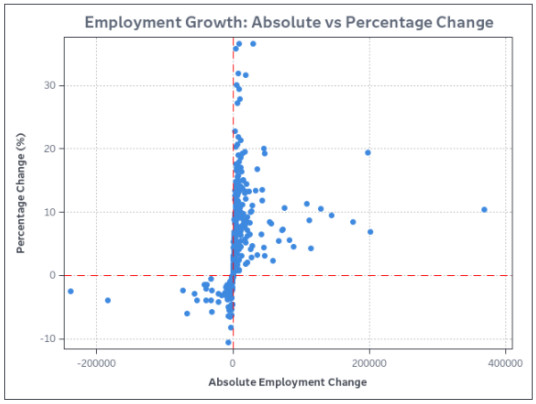

Figure 5. Absolute vs. Percentage Employment Growth Across 396 MSAs

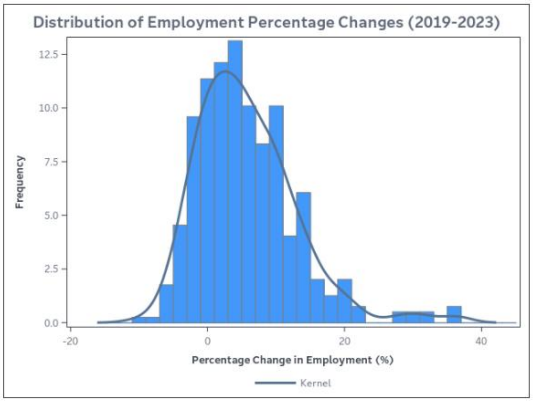

Figure 6. Distribution of Employment Percentage Changes Across 396 MSAs (2019–2023)

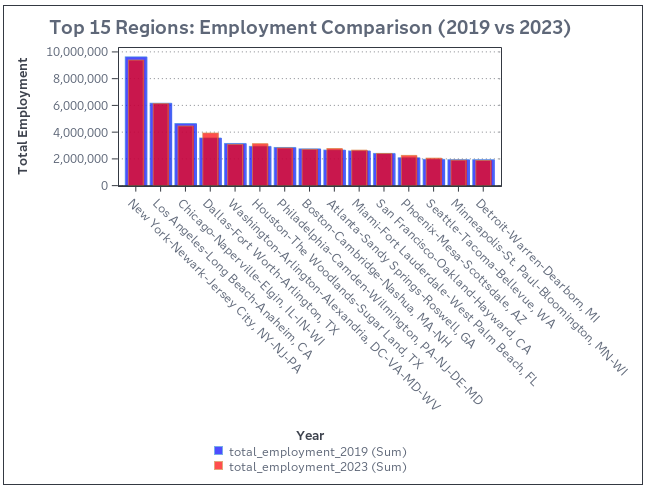

Figure 7. Total Employment Comparison (2019 vs. 2023) for the 15 Largest MSAs